

Current Position

- 2025– **Postdoctoral Researcher**, *Institut Jean Nicod, École Normale Supérieure – PSL*, Paris, France
 Supervisor: Prof. Nicolas Baumard
Fysen Foundation Postdoctoral Study Grant (2025–2026)
 Research areas: cultural evolution; semantic cognition; cognitive ecology; cross-linguistic and cross-cultural modeling; large-scale corpus and LLM-based analysis.

Education

- 2019–2025 **Ph.D., Psychology (Cognitive Neuroscience)**, *Beijing Normal University*, China, Advisor: Prof. Yanchao Bi
 State Key Laboratory of Cognitive Neuroscience and Learning & IDG/McGovern Institute
- 2016–2017 **Visiting Student, Psychology**, *University of California, Berkeley*
- 2014–2019 **B.S., Psychology (Computer Science minor)**, *Central China Normal University*

Research Interests

- Cultural–historical dynamics of collective knowledge
- Cultural evolution and ecological adaptation
- Cross-linguistic universals and cultural variability

Publications and Manuscripts

First / Co-first Author

- 2026 **Fu, Z.**, Chu, Y., Zhang, T., Li, Y., Wang, X., Bi, Y. *Semantic similarity across languages reflects neurocognitive dimensions shaped by climate*. *Nature Communications*, 17(4016).
- 2025 **Fu, Z.***, Chen, H.*, Liu, Z., Song, M., Liu, Z., Bi, Y. *Pathogen stress heightens sensorimotor dimensions in the human collective semantic space*. *Communications Psychology*, 3(2).
- 2023 **Fu, Z.**, Wang, X., Wang, X., Yang, H., Wang, J., Wei, T., Liao, X., Liu, Z., Chen, H., Bi, Y. *Different computational relations in language are captured by distinct brain systems*. *Cerebral Cortex*, 33(4).

Contributing Author

- 2024 Tian, S.*, Chen, L.*, Wang, X., Li, G., **Fu, Z.**, Ji, Y., Lu, J., Wang, X., Shan, S., Bi, Y. *Vision matters for shape representation: Evidence from sculpturing and drawing in the blind*. *Cortex*, 174, 241-255.
- 2023 Tian, S., Chen, Y., **Fu, Z.**, Wang, X., Bi, Y. *Simple shape feature computation across modalities: convergence and divergence between the ventral and dorsal visual streams*. *Cerebral Cortex*, 33(15), 9280-9290.

Talks & Conference Presentations

- 2026 **Text mining reveals a cross-linguistic shared semantic space associated with climate**, Invited online lecture at *Computational Neurolinguistics Forum* (hosted by CUHK), Hong Kong
 Oral
- 2025 **Semantics across the globe**, *International Workshop on Cross-linguistic Databases and Norms*, Shanghai
 Poster

- 2024 *Representing historical cognition in language models*, AI for Psychology Journal Club, Tsinghua University
Oral
- 2024 *Application of language models to semantic cognition*, Lab Meeting, Zhejiang University
Oral
- 2024 *Semantics across the globe*, Rovereto Workshop on Concepts, Actions, and Objects, Trento, Italy
Poster
- 2023 *Ecological drivers of cross-cultural semantic structures*, Conceptual Brains and Cultural Evolution Workshop, Zhuhai
Oral
- 2023 *Pathogen stress heightens sensorimotor dimensions*, Society for the Neurobiology of Language, Marseille
Poster
- 2021 *Graph and vector-embedding models*, Society for the Neurobiology of Language, Online Slam

Research Experience

- 2018–2019 **Research Intern**, State Key Laboratory of Cognitive Neuroscience and Learning, BNU
Advisors: Prof. Xiaoying Wang, Prof. Yanchao Bi
Designed multimodal shape-perception experiments; collected drawings and sculpture data from congenitally blind participants; developed cross-modal computational pipelines.

Mentorship

- 2023–2024 **Undergraduate Thesis Supervision**, Supervised 3 theses (including Outstanding Thesis Award recipient).
- 2022–2024 **Research Training Programs**, Mentor for undergraduate research in the Innovation & Entrepreneurship Training Program.

Professional Service

- Reviewer Imaging Neuroscience; Cognitive Neuropsychology; Humanities and Social Sciences Communications; npj Science of Learning; Discover Artificial Intelligence
- Co-review Nature Human Behaviour.

Honors & Awards

- 2023 Award for Research and Innovation, Beijing Normal University
- 2020–2023 Graduate Academic Award, Beijing Normal University

Technical Skills

- Programming Python; R; MATLAB
- Statistics Multilevel modeling; Network analysis; Time-series modeling; Large-scale corpus analysis
- Methods fMRI; Cross-linguistic modeling; LLM-based semantic analysis; Online behavioral experiments
- Languages Chinese (native), English (fluent), French (beginner)